

Scalability & Future Plan

Date	14 July, 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address(High / Medium / Low)
1	Free-tier cloud deployment memory constraints	Slower response times as user base grows	High
2	Single-worker Flask deployment	Cannot handle a high volume of concurrent credit card applications	Medium
3	Lack of persistent database storage	Applicant data and prediction history are not saved for future model retraining.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Deployed on a single Render free instance handling issue basic traffic.	Migrate to AWS EC2 or Render Pro with Auto-Scaling and Load Balancers to support 10,000+ concurrent requests.
2	Data Storage	Local SQL database with limited capacity	Migrate to scalable cloud storage (Amazon RDS / Azure SQL) with partitioning and backups
3	Performance	Synchronous request processing causing delays under load	Introduce asynchronous processing and a Redis caching layer to reduce latency
4	Security	Basic username / password authentication	Implement OAuth 2.0, data encryption in transit/at rest, and role-based access control

Future Roadmap:

Phase	Planned Feature / Enhancement	Target timeline	Expected Impact
Phase-2	Real-time integration with credit bureau APIs	15/07/2026	Faster and more accurate approval decisions
Phase-3	Mobile app support and multi-language UI	15/07/2026	Wider user reach and improved accessibility
Phase-4	AI-based fraud detection module	15/07/2026	Enhanced security and reduced fraud risk